AI-Based Citizen Grievance Redressal System for Tamil Nadu: A Web Development Perspective

A modern, secure web platform that empowers citizens to submit grievances easily and enables government officers to resolve issues faster using AI. This presentation covers user/officer experiences, core AI features, web architecture, deployment considerations, and the roadmap for phased rollout across Tamil Nadu.

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Introduction: The Need for Smarter Grievance Redressal in Tamil Nadu

Tamil Nadu manages thousands of citizen complaints across urban and rural services daily — from civic complaints to social services. Citizens expect quick acknowledgement, clear status updates, and timely resolution. Government officers need triage, prioritization, and tools to reduce manual overhead while preserving transparency and accountability.

This proposal targets both citizen-facing UX and officer workflows, balancing simplicity with administrative controls and auditability.

Current Challenges: System Limitations & Citizen Frustrations

Fragmented Channels

Complaints arrive via phone, email, in-person, and legacy portals — causing duplication and data loss.

Poor Triage

Critical issues are not consistently prioritized; manual assignment slows response times.

Limited Transparency

Citizens lack clear status updates and escalation paths, reducing trust in government responsiveness.

Officer Overload

High caseloads and repetitive tasks consume time that could be spent on complex casework.

Solution Overview: Introducing the AI-Powered Platform

A responsive web application with two tailored experiences: a citizen portal (mobile-first, Tamil/English) and an officer console (workflows, SLA tracking, analytics). Core AI modules automatically classify, prioritize, route, and suggest responses — reducing manual effort and improving resolution speed.

- Unified intake across channels (web, mobile, IVR, email)
- AI-assisted triage and routing to the correct department or officer
- End-to-end tracking, audit logs, and KPIs for administrators

For Citizens: Intuitive Interface & Seamless Grievance Submission

Simple Multi-Language Submission

Guided form in Tamil and English, with optional voice-to-text input and image/video attachments for evidence.

Instant Acknowledgement & Tracking

Immediate reference number, estimated SLA, and live status updates via SMS, WhatsApp, and the portal.

Clear Escalation and Feedback

Escalate unresolved cases, rate resolutions, and provide feedback that retrains AI models for better future outcomes.

Accessibility-first: WCAG-compliant UI, large fonts, local language support, and low-bandwidth optimizations.

For Officers: AI-Driven Prioritization & Efficient Workflow Management

The officer console centralizes assigned grievances, shows priority flags, SLA timers, and AI-suggested actions. Integrated case notes, evidence viewer, inter-departmental handoffs, and built-in escalation templates make resolution efficient and auditable.

01

Smart Triage

AI classifies category, urgency, and impacted population, routing to the right team automatically.

02

Action Templates

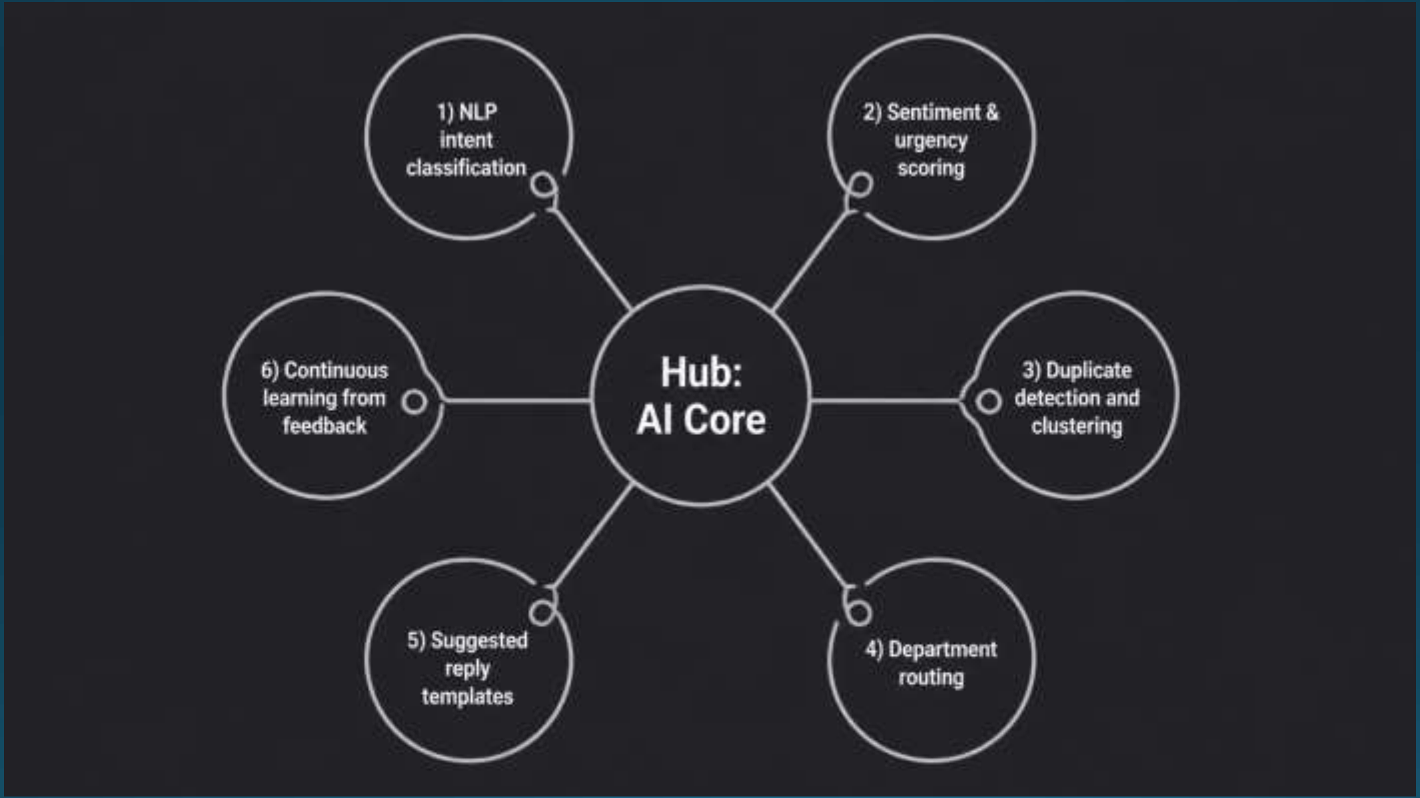
Prefilled response drafts and next-step checklists tailored to complaint type and regulations.

03

Analytics & Dashboards

Real-time KPIs (resolution time, backlog, citizen satisfaction) that inform resource allocation.

Key AI Features: NLP, Sentiment Analysis & Automated Routing



AI capabilities: - NLP to extract intent, location, and service type from free text or voice. - Sentiment + urgency scoring to elevate critical or emotionally charged cases. - Duplicate detection to merge similar complaints and avoid wasted effort. - Automated routing to department/officer with confidence scores and manual override.

Technical Architecture: Web Stack & Scalability Considerations

Proposed stack (scalable, secure, maintainable): - Frontend: React (responsive, Tamil/English i18n), progressive web app (PWA) for offline/low bandwidth use. - Backend: Node.js / Python microservices, REST + GraphQL APIs. - AI: Containerized NLP and ML models served via model server (e.g., FastAPI + TorchServe), with retraining pipelines. - Data: PostgreSQL for transactional data, ElasticSearch for full-text search, object storage for media. - Infrastructure: Kubernetes for autoscaling, CDN for static assets, managed DB, and monitoring (Prometheus/Grafana).

- Security: OAuth2 / SAML for officer SSO, encryption at rest/in transit, role-based access control, audit logs
- Scalability: horizontal scaling, regional failover, and disaster recovery planning

Benefits & Impact: Transparency, Faster Resolution, and Improved Governance

75%

Faster Triage

AI-guided routing reduces manual sorting and queues; anticipated reduction in initial triage time.

50%

Reduced Backlog

Automated clustering and prioritization help focus resources on high-impact cases.

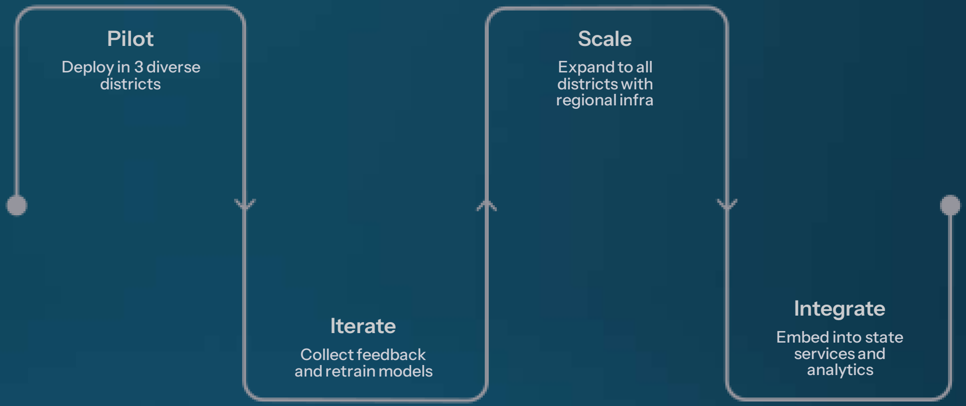
30%

Improved Citizen Satisfaction

Clear status updates and time-bound SLAs increase trust and perceived responsiveness.

Beyond metrics, the platform strengthens civic trust, enables data-driven policy decisions, and creates an auditable record for accountability — aligning with Tamil Nadu's goals for transparent governance.

Future Roadmap: Continuous Improvement & Expansion



Phase 1: Pilot

Deploy in 3 diverse districts, validate UX in Tamil, monitor KPIs, and collect structured feedback.

Phase 2: Iterate

Retrain models with local language nuances, refine workflows, and improve accessibility features.

Phase 3: Scale

Expand to all districts with regional infra, establish SLA governance, and operational runbooks.

Phase 4: Integrate & Evolve

Connect with other state services, open anonymized data for policy insights, and add multilingual AI channels (voice assistants).

Next steps: technical proof-of-concept, stakeholder workshops (citizen groups + officers), and a 6-month pilot agreement.